

Risk measure

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Chapter 2: Coherent risk measure

Plan

- 1 Axiomatic Approach
- 2 Distortion Risk Measures (DRM)
- 3 Spherical and elliptical distributions for the risk factors

- 1 Axiomatic Approach
 - Coherent Risk Measures
 - Incoherency of the VaR
 - Is the coherency a good property?
- 2 Distortion Risk Measures (DRM)
- 3 Spherical and elliptical distributions for the risk factors

Definition of a risk measure

Let $\mathcal{M}$ be a set of random variables defined on a probability space $(\Omega, \mathcal{A}, P)$. One can interpret $\mathcal{M}$ as a set of losses L at fixed horizon. Assume that $\mathcal{M}$ is a convex cone which contains the constants:

$$L_1, L_2 \in \mathcal{M} \text{ and } \lambda \geq 0 \quad \Rightarrow \quad L_1 + L_2 \in \mathcal{M} \text{ and } \lambda L_1 \in \mathcal{M}.$$

Definition (Risk Measure)

A risk measure is an application r from $\mathcal{M}$ to $\mathbb{R}$.

A position of loss L is considered as "unsafe" if $r(L)$ is large. We will interpret $r(L)$ as the cash that must be added to a position of loss L to become acceptable for the regulator (*i.e.* for $r(L) \leq 0$).

Properties of a risk measure

A risk measure r is said to be

invariance by translation if

$$\forall L \in \mathcal{M} \text{ and } \forall \ell \in \mathbb{R}, \quad r(L + \ell) = r(L) + \ell;$$

If $r(L)$ is added to a position of loss L we have a new position of loss $\tilde{L} = L - r(L)$, which is acceptable because $r(\tilde{L}) = 0$.

sub-additive if

$$\forall L_1, L_2 \in \mathcal{M}, \quad r(L_1 + L_2) \leq r(L_1) + r(L_2);$$

The aggregation does not create additional risk, the diversification reduces risk, risk management can be decentralized but this is a controversial property.

Properties of a risk measure (continued)

positively homogeneous if

$$\forall L \in \mathcal{M}, \forall \lambda \geq 0, \quad r(\lambda L) = \lambda r(L);$$

This property is also controversial: for very large λ one could expect $r(\lambda L) > \lambda r(L)$ to penalize the concentration of risk, but this would be in contradiction with subadditivity.

If r is sub-additive and positively homogeneous then it is convex on $\mathcal{M}$:

$$r(\lambda_1 L_1 + (1 - \lambda_1) L_2) \leq \lambda_1 r(L_1) + (1 - \lambda_1) r(L_2).$$

monotone if

$$\forall L_1, L_2 \in \mathcal{M}, \quad L_1 \leq L_2 \text{ a.s.} \Rightarrow r(L_1) \leq r(L_2);$$

If r is sub-additive and positively homogeneous then r is monotone if and only if $r(L) \leq 0 \quad \forall L \leq 0 \text{ a.s.}$

► Proof

Coherent risk measure

A risk measure r is said to be

coherent if it possesses the previous properties (invariance by translation, sub-additivity, positive homogeneity and monotonicity)*.

*Artzner, P., Delbaen, F., Eber, J. M. and Heath, D. (1999) **Coherent measures of risk**. Mathematical finance, 9, 203-228.

Coherency and diversification

For a coherent risk measure, a diversified portfolio is less risky than a concentrated portfolio: if $L_1, \dots, L_n$ are identically distributed (n assets with same loss function), then

$$r\left(\frac{1}{n} \sum_{i=1}^n L_i\right) = \frac{1}{n} r\left(\sum_{i=1}^n L_i\right) \leq \frac{1}{n} \sum_{i=1}^n r(L_i) = r(L_1).$$

More generally, for all $p_i \geq 0$ such that $\sum p_i = 1$ and all $L_1, \dots, L_n$

$$r\left(\sum_{i=1}^n p_i L_i\right) \leq \sum_{i=1}^n r(p_i L_i) = \sum_{i=1}^n p_i r(L_i).$$

What are the coherent risk measures?

Is the coherency always a desirable property?

The VaR is not sub-additive: first example

Let corporate bonds of 100 companies. Assume that the value of each bond is 100 at t and will be equal to :

- 105 at $t+1$ if the company is not in default;
- nothing at $t+1$ if the company is in default.

Assume that the defaults are iid of probability 2%.

The loss L_i of the bond i takes the value

- 100 with probability 2%;
- -5 with probability 98%.

We have

$$L_i = 105Y_i - 5, \quad Y_i = 1_{\{\text{default}\}}.$$

First example (continued)

- The portfolio A (concentrated) contains 100 bonds of Company 1.
- The portfolio B (perfectly diversified) contains 1 bond of each of the 100 companies.

The losses of A and B are:

$$L_A = 100L_1 = 10500Y_1 - 500, \quad L_B = \sum_{i=1}^{100} L_i = 105 \sum_{i=1}^{100} Y_i - 500.$$

At the risk level 5%, the VaR of L_A and L_B are

$$\text{VaR}_A = -500, \quad \text{VaR}_B = 25,$$

noting that $P\{\mathcal{B}(100; 0.02) \leq 5\} \simeq 0.984$ and

$P\{\mathcal{B}(100; 0.02) \leq 4\} \simeq 0.949$. With B the loss is 25 with probability around 5%. With A the probability of losing something is smaller than 5% but when loss happens, all is lost !

The VaR is not sub-additive: second example

- Let L_1 and L_2 independent, such that

$$P(L_i = 10) = 0.009 = 1 - P(L_i = 0).$$

At the risk level 0.01, $\text{VaR}_{L_i} = 0$ and, since $P(L_1 + L_2 = 0) = 0.991^2 = 0.98\dots$, $P(L_1 + L_2 \leq 10) = 1 - 0.009^2 > 0.99$, we have $\text{VaR}_{L_1+L_2} = 10 > \text{VaR}_{L_1} + \text{VaR}_{L_2} = 0$.

- To obtain a version with continuous losses, let us add independent $\mathcal{N}(0, 1)$ random variables: $L_i^* = L_i + \mathcal{N}(0, 1)$. A nontrivial computation shows that

$$\text{VaR}_{L_i^*} \simeq 3.1 \text{ and } \text{VaR}_{L_1^*+L_2^*} \simeq 9.8.$$

We still have $\text{VaR}_{L_1^*+L_2^*} > \text{VaR}_{L_1^*} + \text{VaR}_{L_2^*}$.

The VaR is not sub-additive: third example

Let L_1 and L_2 independent with Pareto distributions, such that

$$P(L_i > x) = \frac{1}{1+x}, \quad x > 0.$$

At the level $\alpha \in (0, 1)$, the VaR of L_i is such that $\frac{1}{1+\text{VaR}} = \alpha$, thus

$$\text{VaR} = \frac{1}{\alpha} - 1.$$

It can be shown that

$$P(L_1 + L_2 > x) = \frac{2}{2+x} + \frac{2\log(1+x)}{(2+x)^2} > P(2L_1 > x), \quad x > 0.$$

Therefore

$$P(L_1 + L_2 > 2\text{VaR}) > \alpha,$$

and thus $\text{VaR}_{L_1+L_2} > 2\text{VaR}$.

What do these examples reveal?

- In Example 1, VaR_{L_A} badly assesses the risk associated to the concentrated portfolio A. This is due to an irrelevant choice of α . Actually a single number (the VaR at a fixed level α) can not resume satisfactorily the entire distribution of a random variable (the loss L_A). In Example 2, the situation is similar, $\text{VaR}(\alpha)$ does not evaluate correctly the risk.
- Example 3 is totally different. Surprisingly, the distribution of the aggregated loss $L_1 + L_2$ has a fatter tail than that of the individual losses $2L_i$. At any risk level α , the VaR correctly indicates that the risk of $L_1 + L_2$ is higher than the risk of $2L_1$. Example 3 reveals that subadditivity may not be a desirable property for losses with very heavy tails, and the VaR does a good job in this framework.

The ES is coherent for integrable losses

The VaR is invariant by translation, positively homogeneous ($F_{\lambda X + \ell}^-(1 - \alpha) = \ell + \lambda F_X^-(1 - \alpha)$) and monotone. The ES, as an average of VaR, shares the same properties. Let us show its sub-additivity *in the continuous case*. For $E|L_i| < \infty$, $i = 1, 2$, we have

$$\begin{aligned} & E \left\{ L_1 1_{L_1 \geq \text{VaR}_{L_1}} + L_2 1_{L_2 \geq \text{VaR}_{L_2}} - (L_1 + L_2) 1_{L_1 + L_2 \geq \text{VaR}_{L_1 + L_2}} \right\} \\ &= \sum_{i=1}^2 E \left\{ L_i \left(1_{L_i \geq \text{VaR}_{L_i}} - 1_{L_1 + L_2 \geq \text{VaR}_{L_1 + L_2}} \right) \right\} \\ &= \sum_{i=1}^2 E (L_i - \text{VaR}_{L_i}) \left(1_{L_i \geq \text{VaR}_{L_i}} - 1_{L_1 + L_2 \geq \text{VaR}_{L_1 + L_2}} \right) \\ &\quad + \text{VaR}_{L_i} E \left(1_{L_i \geq \text{VaR}_{L_i}} - 1_{L_1 + L_2 \geq \text{VaR}_{L_1 + L_2}} \right) \geq 0 \end{aligned}$$

because the last term is null (in the continuous case $P(L_i \geq \text{VaR}_{L_i}) = P(L_1 + L_2 \geq \text{VaR}_{L_1 + L_2}) = \alpha$), and the second last is an expectation of a product of same sign.

Conditional Tail Expectation (CTE)

The Conditional Tail Expectation (CTE) is defined by

$$\text{CTE}_L(\alpha) = E\{L \mid L > \text{VaR}(\alpha)\}.$$

This is the standard definition given in the literature. To avoid conditioning by a zero-probability event, one could modified the definition by setting

$$\text{CTE}_L^*(\alpha) = E\{L \mid L \geq \text{VaR}(\alpha)\}.$$

When L is continuous and strictly increasing,

$$\text{ES}_L(\alpha) = \text{CTE}_L(\alpha) = \text{CTE}_L^*(\alpha).$$

The counterexamples showing that CTE and CTE^* are not coherent thus involve discrete distributions.

CTE is incoherent

Let $X \sim \mathcal{U}_{[0,1]}$ and $Y = (0.95 - X)1_{0 \leq X \leq 0.95} + (1.95 - X)1_{0.95 < X \leq 1}$.

Note that $Y \sim \mathcal{U}_{[0,1]}$ and that $X + Y$ takes the value 0.95 with probability 0.95 and the value 1.95 with probability 0.05.

► Illustration

We have

$$\begin{aligned} & E\{X + Y \mid X + Y > \text{VaR}_{X+Y}(0.1)\} \\ &= E\{X + Y \mid X + Y > 0.95\} = 1.95 \end{aligned}$$

and

$$E\{X \mid X > \text{VaR}_X(0.1)\} = E\{X \mid X > 0.9\} = 0.95,$$

thus

$$\text{CTE}_{X+Y}(0.1) > \text{CTE}_X(0.1) + \text{CTE}_Y(0.1).$$

CTE* is incoherent

Let $U \sim \mathcal{U}_{[-1,1]}$, c_1 and c_2 two constants and

$$X = c_1 1_{U \in [0, 2p]} - c_2 1_{U \notin [0, 2p]}, \quad Y = c_1 1_{U \in [-2p, 0]} - c_2 1_{U \notin [-2p, 0]}$$

with

$$c_1 > 0, \quad c_2 > 0, \quad \text{and} \quad 0 < \frac{\alpha}{2} < p < \alpha < \frac{1}{2}. \quad \text{► Illustration}$$

We have

$$\text{VaR}_X(\alpha) = \text{VaR}_Y(\alpha) = -c_2, \quad \text{VaR}_{X+Y}(\alpha) = c_1 - c_2,$$

$$\text{CTE}_X^*(\alpha) = \text{CTE}_Y^*(\alpha) = EX = pc_1 - (1-p)c_2.$$

Therefore

$$\text{CTE}_{X+Y}^*(\alpha) = c_1 - c_2 > 2\text{CTE}_X^*(\alpha).$$

Comonotonicity

Comonotonicity means **perfect positive dependence**. Comonotonic random variables are increasing functions of the same random variable (they are 'common monotonic').

The vector (L_1, L_2) is said to be comonotonic

iff L_1 and L_2 are increasing functions of a same random variable Z
iff

$$\forall \omega, \omega^* \in \Omega, \quad \{L_1(\omega) - L_1(\omega^*)\} \{L_2(\omega) - L_2(\omega^*)\} \geq 0$$

iff

$$(L_1, L_2) \stackrel{\mathcal{L}}{=} (F_{L_1}^-(U), F_{L_2}^-(U)), \quad U \sim \mathcal{U}_{[0,1]}.$$

Intuitively, L_1 and L_2 **always move in the same direction**.

Additivity for comonotonic risks

It is said that r is additive for comonotonic risks/losses iff

$$(L_1, L_2) \text{ comonotonic} \Rightarrow r(L_1 + L_2) = r(L_1) + r(L_2).$$

Diversification does not reduce the risk when the risk factors are comonotonic.

Lemma

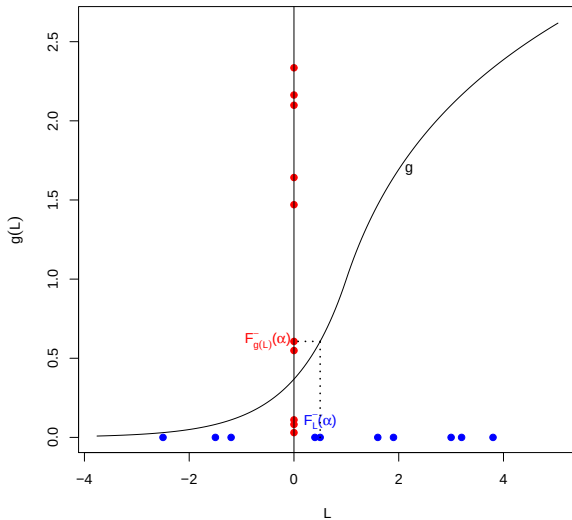
If g is continuous and strictly increasing, and $\alpha \in (0, 1)$ then

$F_{g(L)}^-(\alpha) = g\{F_L^-(\alpha)\}$ (equivariance property, see

<https://arxiv.org/pdf/1004.0533.pdf>).

Proof: $\forall x, \quad F_{g(L)}^-(\alpha) \leq x \Leftrightarrow \alpha \leq F_{g(L)}(x) = F_L(g^{-1}(x)) \Leftrightarrow F_L^-(\alpha) \leq g^{-1}(x) \Leftrightarrow g\{F_L^-(\alpha)\} \leq x$

Equivariance of the quantile (illustration of the lemma)



VaR and ES for comonotonic risks

Properties

VaR and ES are additive for comonotonic risks.

Proof in the case L_1 and L_2 continuous, strictly increasing and comonotonic: The quantile functions $F_{L_1}^{-1}$ and $F_{L_2}^{-1}$ are strictly increasing from $(0, 1)$ to $\mathbb{R}$, and we have

$$\begin{aligned}\text{VaR}_{L_1+L_2}(\alpha) &= \text{VaR}_{(F_{L_1}^{-1}+F_{L_2}^{-1})(U)}(\alpha) \\ &= (F_{L_1}^{-1} + F_{L_2}^{-1})\{F_U^{-1}(1-\alpha)\} \\ &= F_{L_1}^{-1}(1-\alpha) + F_{L_2}^{-1}(1-\alpha) = \text{VaR}_{L_1}(\alpha) + \text{VaR}_{L_2}(\alpha),\end{aligned}$$

using $L_1 + L_2 \sim (F_{L_1}^{-1} + F_{L_2}^{-1})(U)$, the previous lemma and $F_U^{-1}(1-\alpha) = 1-\alpha$.

Diversification can amplify solvency problems

Let 3 events ω_i with same probability $1/3$ and the losses

	L_1	L_2	$2L_1$	$L_1 + L_2$
ω_1	150	10	300	160
ω_2	10	150	20	160
ω_3	-200	-200	-400	-400

For L_1 (or L_2) to be acceptable with probability $2/3$, a capital requirement of 10 is needed. For the loss $L_1 + L_2$ be acceptable, a capital requirement of 160 is needed. The interactions deteriorate the global solvency because a small loss is associated with a large loss.

The VaR takes into account the solvency deterioration :

$$\text{VaR}(2L_1) < \text{VaR}(L_1 + L_2),$$

which is not possible with a coherent measure.

Non existence of coherent risk measures if $\mathcal{M}$ contains heavy-tailed variables

Suppose that $\mathcal{M}$ contains variables with the so-called α_0 -stable distribution, for which

$$P(|X_i| > x) = O(x^{-\alpha_0}) \quad \text{as } x \rightarrow \infty$$

with $\alpha_0 \in (0, 2)$. We have, for an i.i.d. sequence of α_0 -stable variables (X_i) ,

$$n^{-1/\alpha_0} \sum_{i=1}^n X_i \stackrel{d}{=} X_1.$$

If the risk is positively homogeneous and only depends of the loss distribution,

$$r\left(\sum_{i=1}^n X_i\right) = n^{1/\alpha_0} r(X_1) > nr(X_1) = \sum_{i=1}^n r(X_i), \text{ for } \alpha_0 < 1.$$

The VaR is incoherent. Is there any problem?

- In particular situations (the 100 corporate bonds example), the sub-additivity of the VaR leads to misjudge the risk, but this is mainly due to an irrelevant choice of α .
- If $\mathcal{M}$ contains "extreme" losses, as in the Pareto example, diversification may increase the risk, and then the over-additivity of the VaR is called for (ES is useless since $ES(\alpha) = \infty$).
- Over-additivity occurs when the interactions alter the global solvency.
- Over-additivity occurs for α -stable distributions with $\alpha < 1$, and this example reveals that the term "coherent" is misleading.
- Imagine that several people collect mushrooms in a forest. Is it less risky to eat an omelette with the mushrooms of a common basket than with your own mushrooms?

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- 2 Distortion Risk Measures (DRM)
 - Definition of a DRM
 - Other interpretation of a DRM
 - Parametric DRM
- 3 Spherical and elliptical distributions for the risk factors

DRM

Let an **integrable** loss with a **continuous and strictly increasing** cdf F . Note that

$$\text{ES}(\alpha) = \int F^{-1}(1-u) \frac{1}{\alpha} \mathbf{1}_{[0,\alpha]}(u) du$$

assigns weight uniformly over the worst $[0, \alpha]$ losses. A different weighting scheme can be introduced.

A DRM (Distortion Risk Measure)

is defined by

$$r(F, G) = \int_0^1 F^{-1}(1-u) dG(u),$$

where G is a cdf over $[0, 1]$ called **distortion function**.

Distortion: probability law on the confidence levels

$$r(F, G) = \int_0^1 F^{-1}(1-u) dG(u)$$

Examples:

- $ES(\alpha)$: the distortion G uniform law on $[0, \alpha]$.
- $VaR(\alpha)$: G Dirac mass at α .
- $E(L) = E\{F^{-1}(U)\} = \int_0^1 F^{-1}(1-u) du$: $G = U$ uniform on $[0, 1]$.

Interpretation in terms of optimism or pessimism in the face of risk:

If G admits a density g which is increasing over $[0, 1]$

($g'(y) = G''(y) \geq 0$, G convex), large risks (small α) are not much taken into account (optimistic behavior).

Conversely, if g is decreasing (G concave), extreme risks are heavily weighted, which corresponds to pessimism/prudential attitude towards risks.

DRM as the expectation of a distorted loss

It is well known that for X integrable with density

$$EX = \int_0^{+\infty} P\{X > x\} dx - \int_{-\infty}^0 P(X < x) dx$$

► Proof

It can be shown that, when F admits a density,

$$r(F; G) = \int_0^{+\infty} G\{S(x)\} dx - \int_{-\infty}^0 [1 - G\{S(x)\}] dx$$

► Proof

where $S(x) = 1 - F(x) = P[L > x]$ (survival function).

The DRM satisfies $r(F, G) = E(L^*)$,

where L^* has survival function $G(S(x))$.

After distortion by G , the loss L becomes L^* .

For $G(x) = x$ we retrieve $E(L)$.

Note that L^* stochastically dominates L if G is concave ($S \leq G(S)$).

Proportional hazard DRM

Taking $G_p(u) = u^p$ with $p > 0$,

$$\begin{aligned} r(F, G) &= \int_0^1 F^{-1}(1-u) dG_p(u) \\ &= \int_0^1 F^{-1}(1-u) p u^{p-1} du = E(L_p^*), \end{aligned}$$

where $P(L_p^* > x) = \{P(L > x)\}^p$.

Note that G_p is concave if $p < 1$ (large weights for large risks) and G_p is convex if $p > 1$ (optimism).

The hazard functions[†] of L_p^* and L are proportional:

$$-\frac{\partial \log P(L_p^* > x)}{\partial x} = -\frac{\partial \log P(L > x)}{\partial x} p.$$

[†] see Survival Analysis or Duration Analysis:

$$\lim_{\delta \rightarrow 0^+} \frac{P(x < L < x + \delta \mid L > x)}{\delta} = -\frac{\partial \log P(L > x)}{\partial x}.$$

Exponential DRM

Taking $G_p(u) = \frac{1-e^{-pu}}{1-e^{-p}}$ with $p > 0$,

$$r(F, G) = \int_0^1 F^{-1}(1-u) \frac{pe^{-pu}}{1-e^{-p}} du,$$

Note $G_p(u)$ is concave.

It can be shown that a DRM with G concave is coherent.

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- 3 Spherical and elliptical distributions for the risk factors
 - Spherical laws
 - Elliptical distributions
 - Sub-additivity of the VaR for elliptical risk factors

An interesting property of Gaussian losses

Suppose

$$L \stackrel{d}{=} \boldsymbol{\lambda}' \mathbf{X} + c,$$

where $\boldsymbol{\lambda} \in \mathbb{R}^d$ and $c \in \mathbb{R}$ are constants and $\mathbf{X}$ is a random vector. If $\mathbf{X} \sim \mathcal{N}(\mathbf{m}, \boldsymbol{\Sigma})$ then

$$\text{VaR}(\alpha) = \boldsymbol{\lambda}' \mathbf{m} + c + \sqrt{\boldsymbol{\lambda}' \boldsymbol{\Sigma} \boldsymbol{\lambda}} \Phi^{-1}(1 - \alpha).$$

The assumption of Gaussianity of financial returns is however unrealistic (heavy-tailed distributions, frequent extreme values).

We will extend the formula for a class of multivariate distributions that does not constraint the tail fatness but requires a property of symmetry in all directions.

Spherical distributions

Definition

The vector $\mathbf{X} = (X_1, \dots, X_d)'$ has a spherical distribution iff either

- ① $\mathbf{P}\mathbf{X} \stackrel{\mathcal{L}}{=} \mathbf{X}$ for any orthogonal transformation $\mathbf{P}$ ($\mathbf{P}\mathbf{P}' = \mathbf{P}'\mathbf{P} = \mathbf{I}_d$)
- ② $\varphi_{\mathbf{X}}(\mathbf{t}) := Ee^{i\mathbf{t}'\mathbf{X}} = \psi(\|\mathbf{t}\|)$ where $\|\mathbf{t}\|^2 = \sum_{i=1}^d t_i^2$. The application $\psi: \mathbb{R} \rightarrow \mathbb{R}$ is called the characteristic generator.
- ③ $\forall \boldsymbol{\lambda} \in \mathbb{R}^d, \boldsymbol{\lambda}'\mathbf{X} \stackrel{\mathcal{L}}{=} \|\boldsymbol{\lambda}\|X_1$.

► Proof that the 3 conditions are equivalent

Example

The Gaussian vector $\mathbf{X} \sim \mathcal{N}(0, \mathbf{I}_d)$ has a spherical distribution with generator $\psi(t) = e^{-t^2/2}$:

$$\varphi_{\mathbf{X}}(\mathbf{t}) = e^{-\mathbf{t}'\mathbf{t}/2}.$$

Spherical distribution and uniform law on the sphere

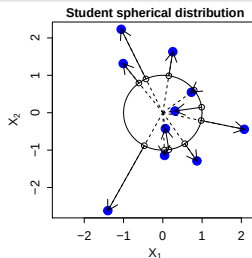
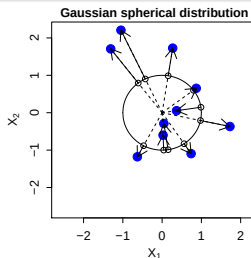
Proposition

The vector $\mathbf{X}$ in $\mathbb{R}^d$ has a spherical distribution iff

$$\mathbf{X} \stackrel{\mathcal{L}}{=} R\mathbf{S},$$

where R is a positive r.v., independent of $\mathbf{S}$, and $\mathbf{S}$ has a uniform law on the sphere $\mathcal{S} = \{\mathbf{s} \in \mathbb{R}^d : \|\mathbf{s}\| = 1\}$.

► Proof



Spherical Gaussian variable in the form RS

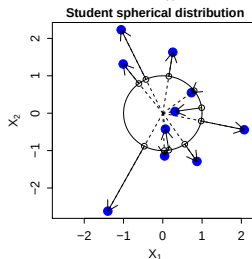
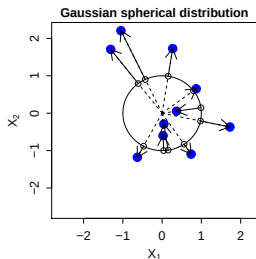
If $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ then

$$\mathbf{X} = \|\mathbf{X}\| \frac{\mathbf{X}}{\|\mathbf{X}\|}$$

where

$$\frac{\mathbf{X}}{\|\mathbf{X}\|} \sim F_{\mathcal{S}} \text{ is independent of } \|\mathbf{X}\|^2 \sim \chi_d^2.$$

It follows that if $\mathbf{S} \sim F_{\mathcal{S}}$ then $E\mathbf{S} = \mathbf{0}$ and $\text{Var}\mathbf{S} = \frac{1}{d}\mathbf{I}_d$.



Student multivariate distribution (multivariate t -distribution)

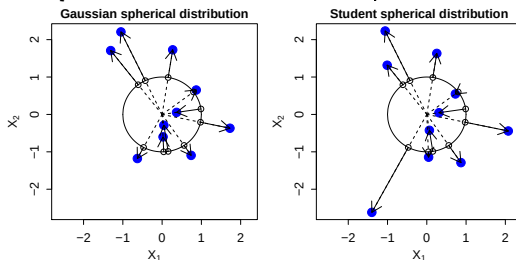
If $V \sim \chi_v^2$, $Z \sim \mathcal{N}(0, 1)$, if V and Z are independent then

$$\frac{Z}{\sqrt{V/v}} \sim \text{St}_v.$$

More generally, if $v/W \sim \chi_v^2$, $\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, W and $\mathbf{Z}$ independent, then

$$\sqrt{W}\mathbf{Z} \sim \text{multivariate } t\text{-distribution } t_d(v, \boldsymbol{\mu}, \boldsymbol{\Sigma}).$$

When $\boldsymbol{\Sigma} = \mathbf{I}_d$ and $\boldsymbol{\mu} = 0$, the distribution is spherical.



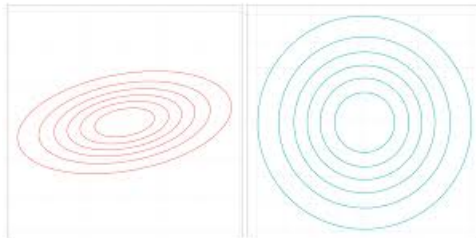
Definition of an elliptical law

Definition

Vector $\mathbf{X}$ has an elliptical law iff

$$\mathbf{X} \stackrel{\mathcal{L}}{=} \boldsymbol{\mu} + \mathbf{A}\mathbf{Y},$$

where $\mathbf{Y}$ has a spherical distribution with generator ψ in $\mathbb{R}^k$, $\boldsymbol{\mu} \in \mathbb{R}^d$, $\mathbf{A}$ $d \times k$. Notation: $\mathbf{X} \sim E_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \psi)$, $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}'$.



Level lines of elliptical (left) and spherical (right) bivariate distributions

Properties of an elliptical law

Properties

If $\mathbf{X} \sim E_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \psi)$, $\mathbf{B} \ell \times d$ and $\mathbf{b} \in \mathbb{R}^\ell$, then
 $\mathbf{BX} + \mathbf{b} \sim E_\ell(\mathbf{B}\boldsymbol{\mu} + \mathbf{b}, \mathbf{B}\boldsymbol{\Sigma}\mathbf{B}', \psi)$. If $\boldsymbol{\lambda} \in \mathbb{R}^d$,

$$\boldsymbol{\lambda}'\mathbf{X} \stackrel{\mathcal{L}}{=} \boldsymbol{\lambda}'\boldsymbol{\mu} + \|\boldsymbol{\lambda}'\mathbf{A}\| Y_1, \quad \text{VaR}_{\boldsymbol{\lambda}'\mathbf{X}}(\alpha) = \boldsymbol{\lambda}'\boldsymbol{\mu} + \|\boldsymbol{\lambda}'\mathbf{A}\| \text{VaR}_{Y_1}(\alpha).$$

VaR is sub-additive for elliptical risks

Proposition

Let $\mathbf{X} \sim E_d(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \psi)$ and $\mathcal{M} = \{L: L = \lambda_0 + \sum_{i=1}^d \lambda_i X_i, \quad \lambda_i \in \mathbb{R}\}$. For all L_1 and L_2 in $\mathcal{M}$ and all $\alpha \in [0, 1/2]$

$$\text{VaR}_{L_1+L_2}(\alpha) \leq \text{VaR}_{L_1}(\alpha) + \text{VaR}_{L_2}(\alpha).$$

Proof: For all $L \in \mathcal{M}$, with obvious notations

$$L = \lambda_0 + \boldsymbol{\lambda}'\mathbf{X} = \lambda_0 + \boldsymbol{\lambda}'\boldsymbol{\mu} + \boldsymbol{\lambda}'\mathbf{A}\mathbf{Y} \stackrel{\mathcal{L}}{=} c_0 + \|\boldsymbol{\lambda}'\mathbf{A}\| Y_1.$$

Noting that the components of a spherical distribution are symmetric, $\text{VaR}_{Y_1}(\alpha) \geq 0$ when $\alpha \leq 1/2$, and

$$\begin{aligned} \text{VaR}_{L+L^*}(\alpha) &= c_0 + c_0^* + \|(\boldsymbol{\lambda} + \boldsymbol{\lambda}^*)'\mathbf{A}\| \text{VaR}_{Y_1}(\alpha) \\ &\leq c_0 + \|\boldsymbol{\lambda}'\mathbf{A}\| \text{VaR}_{Y_1}(\alpha) + c_0^* + \|\boldsymbol{\lambda}^*\mathbf{A}\| \text{VaR}_{Y_1}(\alpha). \end{aligned}$$

End of Chapter 2 ☺ !

r is monotone if and only if $r(L) \leq 0 \ \forall L \leq 0$ a.s.

Suppose r is subadditive and positively homogeneous.

- If r is monotone and $L \leq 0$ a.s., then $r(L) \leq r(0) = 0$. This shows the "only if" part.
- If $L_1 \leq L_2$ and $r(L_1 - L_2) \leq 0$ then

$$r(L_1) = r(L_1 - L_2 + L_2) \leq r(L_1 - L_2) + r(L_2) \leq r(L_2).$$

This show the "if" part.

Mean and survival function

Let X be a random variable with density f . The function $S(x) = P(X > x)$ is called the **survival function**.

If $X \geq 0$, by Fubini

$$\int_0^\infty S(x) dx = \int_0^\infty \left(\int_x^\infty f(t) dt \right) dx = \int_0^\infty \left(\int_0^t f(t) dx \right) dt = \int_0^\infty tf(t) dt = EX.$$

For X integrable, we have

$$\begin{aligned} EX &= EX^+ - EX^- = \int_0^\infty S(x) dx - \int_0^\infty P(X^- > x) dx \\ &= \int_0^\infty S(x) dx - \int_{-\infty}^0 P(X < x) dx. \end{aligned}$$

$r(F, G) = E(L^*)$ where L^* has survival function $G(S(x))$

Proof: By integration by parts, and changes of variables $v = 1 - u$ and $x = F^{-1}(v)$, we have

$$\begin{aligned}
 & r(F, G) \\
 &= \int_0^{1-F(0)} F^{-1}(1-u) dG(u) + \int_{1-F(0)}^1 F^{-1}(1-u) d[G(u) - 1] \\
 &= - \int_0^{1-F(0)} G(u) dF^{-1}(1-u) - \int_{1-F(0)}^1 [G(u) - 1] dF^{-1}(1-u) \\
 &= \int_{F(0)}^1 G(1-v) dF^{-1}(v) + \int_0^{F(0)} [G(1-v) - 1] dF^{-1}(v) \\
 &= \int_0^\infty G(S(x)) dx - \int_{-\infty}^0 [1 - G\{S(x)\}] dx \\
 &= E(L^*).
 \end{aligned}$$

Proof of the equivalence between the 3 Definitions

- 1) $\Rightarrow$ 2): If $\mathbf{t}$ and $\mathbf{s}$ such that $\|\mathbf{t}\| = \|\mathbf{s}\|$ (on the same sphere) then there exists a rotation $\mathbf{P}$ such that $\mathbf{t} = \mathbf{P}\mathbf{s}$. We thus have

$$\varphi_X(\mathbf{t}) = \varphi_{\mathbf{P}\mathbf{X}}(\mathbf{t}) = Ee^{i\mathbf{t}'\mathbf{P}\mathbf{X}} = \varphi_X(\mathbf{P}'\mathbf{t}) = \varphi_X(\mathbf{s}).$$

- 2) $\Rightarrow$ 3): Letting $\mathbf{e}_1$ the 1st column of $\mathbf{I}_d$, we have

$$\varphi_{X_1}(t_1) = Ee^{it_1X_1} = Ee^{it_1\mathbf{e}_1'X} = \varphi_X(t_1\mathbf{e}_1) = \psi(|t_1|).$$

This entails

$$\begin{aligned}\varphi_{\lambda'X}(t_1) &= Ee^{it_1\lambda'X} = \varphi_X(t_1\lambda) = \psi(|t_1|\|\lambda\|) \\ &= \varphi_{X_1}(t_1\|\lambda\|) = \varphi_{\|\lambda\|X_1}(t_1).\end{aligned}$$

- 3) $\Rightarrow$ 1): For any orthogonal matrix $\mathbf{P}$, we have

$$\varphi_{\mathbf{P}\mathbf{X}}(\mathbf{t}) = Ee^{i(\mathbf{P}'\mathbf{t})'X} = Ee^{i\|\mathbf{P}'\mathbf{t}\|X_1} = Ee^{i\|\mathbf{t}\|X_1} = Ee^{i\mathbf{t}'X} = \varphi_X(\mathbf{t}).$$

Proof that $\mathbf{X} = \mathbf{R}\mathbf{S}$ implies $\mathbf{X}$ spherical

- ① Suppose $\mathbf{X} = \mathbf{R}\mathbf{S}$ with $R \geq 0$ independent of $\mathbf{S}$ and $\mathbf{S}$ uniformly distributed on $\mathcal{S}$. Let ψ_d the generator of the spherical law of $\mathbf{S}$. We have

$$\begin{aligned}\varphi_{\mathbf{R}\mathbf{S}}(\mathbf{t}) &= E\{E(e^{i\mathbf{t}\mathbf{R}\mathbf{S}} \mid R)\} = \int \psi_d(\|\mathbf{r}\mathbf{t}\|) dP_R(r) \\ &= \int \psi_d(r\|\mathbf{t}\|) dP_R(r) = \psi(\|\mathbf{t}\|)\end{aligned}$$

hence $\mathbf{R}\mathbf{S}$ has a spherical law.

Proof that $\mathbf{X}$ spherical implies $\mathbf{X} = R\mathbf{S}$

- ① Suppose that $\mathbf{X}$ has spherical law of generator ψ . For all $\mathbf{s} \in \mathcal{S}$, we have

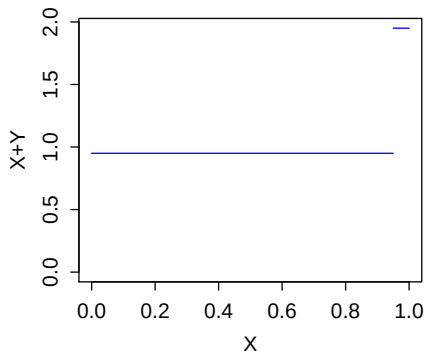
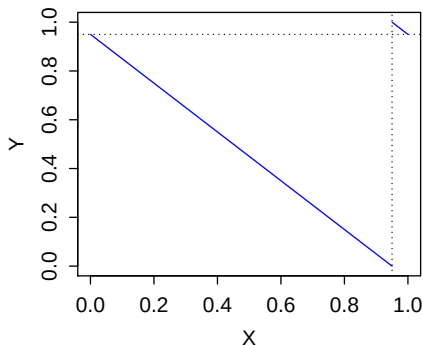
$$\psi(\|\mathbf{t}\|) = \varphi_{\mathbf{X}}(\mathbf{t}) = \varphi_{\mathbf{X}}(\|\mathbf{t}\|\mathbf{s}).$$

Letting $F_{\mathcal{S}}$ the uniform law on $\mathcal{S}$,

$$\begin{aligned} \psi(\|\mathbf{t}\|) &= \int_{\mathcal{S}} \varphi_{\mathbf{X}}(\|\mathbf{t}\|\mathbf{s}) dF_{\mathcal{S}}(\mathbf{s}) = E \int_{\mathcal{S}} e^{i\|\mathbf{t}\|\mathbf{X}'\mathbf{s}} dF_{\mathcal{S}}(\mathbf{s}) \\ &= E \psi_d(\|\mathbf{t}\|\|\mathbf{X}\|) = \int \psi_d(r\|\mathbf{t}\|) dF_{\|\mathbf{X}\|}(r), \end{aligned}$$

which is the characteristic function of $R\mathbf{S}$ where R and $\mathbf{S}$ are independent, $R \sim \|\mathbf{X}\|$ and $\mathbf{S} \sim F_{\mathcal{S}}$.

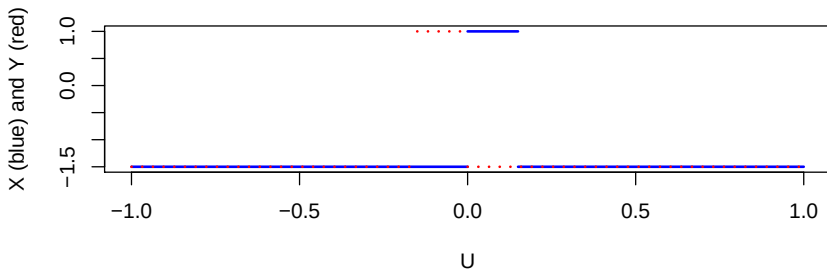
$$X \sim U[0, 1], \quad Y = (0.95 - X)1_{[0, 0.95]}(X) + (1.95 - X)1_{(0.95, 1]}(X)$$



X and Y are uniformly distributed on $[0, 1]$, $X + Y = 0.95$ with probability 0.95 and $X + Y = 1.95$ with probability 0.05

◀ Return

$$U \sim \mathcal{U}[-1, 1], \quad X = c_1 1_{[0, 2p]}(U) - c_2 1_{[0, 2p]^c}(U), \quad Y = c_1 1_{[-2p, 0]}(U) - c_2 1_{[-2p, 0]^c}(U)$$



X and Y take the values c_1 and $-c_2$ with probability p and $1 - p \geq 1 - \alpha$. $X + Y$ takes the values $c_1 - c_2$ and $-2c_2$ with probability $2p$ and $1 - 2p < 1 - \alpha$.